

Machine Learning Term Project Proposal

3.1 Project Title

Papers, Please

3.2 Group Information

- Team Leader: Connor Kress - ckress@ufl.edu
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3.3 Problem Description

- The problem our team is trying to solve is how camera captures can identify documents and locate the corners of the documents.
- Many people could use software to help digitize documents easier, especially students and banks. Students who wish to submit their paperwork online may not always have access to a scanner but will usually be able to use their phone cameras. A picture can work however the student may only wish to submit a full-screen image of the document. Image recognition software can help with this task by making it easier to find and focus the document using the camera. Banks also try to digitize documents, especially paper checks. However, many people wish to submit written checks to the bank and expect a fast response. Image recognition software can help banks deal with the large amount of checks they receive. Once the corners are located, the document can be cropped and corrected to produce a properly aligned image.
- This project will be treated as a multi-output regression task using deep learning. The model will predict eight continuous values representing the horizontal and vertical coordinates of the document's four corners.

3.4 Dataset Description

The project will use the SmartDoc 2015 Challenge 1 Dataset:

<https://github.com/jchazalon/smartdoc15-ch1-dataset>

That dataset contains 24,889 annotated video frames from 150 video clips captured at 1920x1080 pixels. It was created using 30 printed documents from six document types:

- Datasheets
- Letters
- Magazines
- Papers
- Patents
- Tax Forms

Each type contains five different documents, and each document was recorded in five different background scenarios. The frames include realistic conditions like distortion, motion blur, different lighting, and partial obstruction

Each sample consists of an image frame and metadata containing the coordinates of the document's top-left, bottom-left, bottom-right, and top-right corners. The image's numerical pixel values will be used as the model's inputs, while the eight corner-coordinate values will serve as the regression targets. The dataset contains more than 1,000 labeled samples, varied document layouts and capture conditions, and supports a real-world document detection application.

Since there are many different kinds of documents that could be scanned, it is important to train the deep learning model on different kinds of documents. Thus, the six document types above are the different types of features that we will be using for our models. The target variable is classification of the document into one of the six types.

3.5 Initial Project Plan

- **Data cleaning and preprocessing**
 - Check for missing or corrupted images and annotations.
 - Sample frames from each video so we do not use too many nearly identical images.
 - Resize the images to the same size and normalize the pixel values.
 - Convert the four corner locations into eight normalized values between 0 and 1.
 - Use basic data augmentation, such as changes in brightness, blur, rotation, and perspective.
- **Exploratory data analysis and visualizations**
 - Display sample images with the labeled document corners drawn on them.
 - Look at the distribution of corner coordinates and document sizes.
 - Compare images from different backgrounds and video sequences.

- Check how much overlap exists between nearby video frames.
- **Machine learning algorithms**
 - Ridge regression using resized and flattened images.
 - A multilayer perceptron, or MLP.
 - A convolutional neural network, or CNN, built in PyTorch.
 - If there is time, we could test a second CNN with more layers or batch normalization.
- **Cross-validation and hyperparameter tuning**
 - Split the data by video instead of by individual frame to prevent data leakage.
 - Use grouped cross-validation for the simpler models.
 - Use a training, validation, and test split for the CNN.
 - Tune learning rate, batch size, number of layers, dropout, and weight decay.
 - Use early stopping based on validation loss.
- **Evaluation metrics**
 - Mean corner error.
 - Normalized mean corner error.
 - Intersection over union, or IoU, between the predicted and actual document area.
 - Percentage of predictions above a chosen IoU threshold.
 - Average inference time per image.
 - Visual examples of correct predictions and failure cases.

3.6 Contribution Table

Name	Tasks
Connor Kress	Cloud compute management (GPUs for deep learning), results interpretation, and helping with model implementation / tuning
Joshua Rivera	Problem definition, exploratory data analysis / visualization, and background research, Presentation preparation
Cole Morgan	Data cleaning and preprocessing and feature engineering (cropping images, color channel management, etc.)
Samy Katibi	Dataset collection/documentation, exploratory data analysis/visualization, and presentation preparation
Vinh Nguyen	Model implementation; cross-validation and hyperparameter tuning; and model evaluation and comparison

